

PORTALEGRE, Portugal

WMO: 085710

Lat: 39.2942N

Lon: 7.4214W

Elev: 598

StdP: 94.35

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	1.5	2.7	-10.1	1.7	7.9	-7.4	2.2	8.8	10.5	7.0	9.5	7.3	4.7	320	0.454

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	12.4	35.7	18.5	34.0	18.1	32.3	17.6	19.9	31.5	19.2	30.1	18.6	29.1	2.6	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
16.8	12.9	20.5	16.1	12.3	20.1	15.4	11.7	19.8	59.4	31.7	56.9	30.3	54.9	29.2	22.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
9.1	8.0	7.1	-0.6	38.7	1.5	1.3	-1.7	39.7	-2.6	40.5	-3.4	41.2	-4.5	42.2		
			-2.9	20.8	1.9	0.9	-4.2	21.5	-5.3	22.0	-6.4	22.6	-7.8	23.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	15.8	8.6	9.5	11.8	13.2	16.9	21.5	23.9	24.4	21.5	17.1	11.5	9.5
	DBStd	6.84	2.77	3.12	3.77	4.30	4.72	4.80	4.23	4.07	4.14	4.00	3.24	2.62
	HDD10.0	200	59	43	23	13	1	0	0	0	0	1	19	41
	HDD18.3	1592	301	247	206	164	86	20	2	1	14	73	204	273
	CDD10.0	2329	16	29	78	109	216	346	432	447	345	220	66	25
	CDD18.3	682	0	0	2	11	43	116	177	189	110	33	1	0
	CDH23.3	6791	0	0	3	41	329	1214	2033	2096	917	158	0	0
	CDH26.7	2916	0	0	0	2	90	507	963	991	329	34	0	0
Wind	WSAvg	3.3	3.3	3.5	3.7	3.5	3.3	3.2	3.3	3.2	2.9	3.1	3.4	3.4
Precipitation	PrecAvg	881	122	107	76	79	63	35	7	8	43	99	120	121
	PrecMax	1448	436	266	280	291	212	183	58	47	118	298	386	392
	PrecMin	602	0	5	0	2	0	0	0	0	1	2	1	13
	PrecStd	209	100	79	66	55	47	39	11	11	32	74	93	99
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	17.2	18.9	23.3	26.8	31.9	36.6	37.6	37.9	34.6	30.2	21.2	17.7
		MCWB	10.1	11.4	12.7	15.4	16.9	19.0	19.0	19.0	17.9	15.3	13.9	11.4
	2%	DB	14.8	17.0	21.1	24.4	29.1	33.8	35.6	35.7	32.3	27.1	19.0	15.7
		MCWB	9.0	10.9	12.0	14.6	16.9	18.1	18.5	18.5	17.1	14.8	12.9	10.6
	5%	DB	13.4	15.4	19.4	22.2	26.9	31.8	33.8	33.8	30.3	24.9	17.4	14.2
		MCWB	9.0	9.8	11.4	13.5	15.8	17.7	18.1	18.1	16.6	14.2	12.2	10.2
	10%	DB	12.2	13.9	17.6	20.3	24.8	29.6	31.9	31.8	28.3	22.8	15.9	13.0
		MCWB	8.8	9.1	10.6	12.6	15.0	17.0	17.5	17.5	16.0	13.9	11.7	10.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.8	13.4	14.5	16.7	19.2	20.5	21.1	20.7	19.3	18.2	16.2	13.6
		MCDB	13.7	16.5	20.5	24.2	28.2	32.3	33.3	33.7	29.7	22.8	17.8	14.2
	2%	WB	11.7	12.1	13.2	15.3	17.9	19.4	19.9	19.8	18.5	17.0	14.9	12.7
		MCDB	12.6	14.3	18.2	22.6	26.6	30.5	31.6	31.6	27.7	21.7	16.6	13.7
	5%	WB	10.8	11.2	12.3	14.2	16.8	18.6	19.1	19.1	17.8	16.2	14.0	11.8
		MCDB	11.8	13.6	16.7	21.0	25.1	29.1	30.2	30.3	26.4	20.7	16.0	12.9
	10%	WB	10.0	10.2	11.5	13.1	15.8	17.8	18.3	18.4	17.1	15.5	13.0	10.9
		MCDB	11.2	12.7	15.6	19.1	22.9	27.8	29.2	29.1	24.9	19.7	14.6	12.2
Mean Daily Temperature Range	5% DB	MDBR	5.1	6.2	7.2	8.2	9.4	11.4	12.9	12.4	9.9	7.0	5.4	4.8
		MCDBR	6.3	7.5	8.5	9.4	11.0	12.1	12.6	12.2	11.0	9.0	6.4	5.7
		MCWBR	4.6	4.8	5.1	5.2	5.4	5.2	5.0	4.8	4.5	4.3	4.2	4.2
	5% WB	MCDBR	4.8	6.3	7.6	9.1	10.5	11.8	13.3	12.5	10.3	7.1	5.2	4.5
		MCWBR	4.1	4.5	5.0	5.3	5.6	5.3	5.3	5.1	4.5	3.9	4.0	4.1
Clear-Sky Solar Irradiance	taub	0.298	0.314	0.342	0.344	0.351	0.354	0.337	0.352	0.345	0.334	0.310	0.298	
	taud	2.523	2.481	2.409	2.416	2.422	2.429	2.447	2.409	2.454	2.512	2.532	2.530	
	Ebn at Noon	882	913	915	932	927	920	933	912	900	877	860	854	
	Edh at Noon	75	90	107	113	115	114	111	113	101	86	74	70	
All-Sky Solar Radiation	RadAvg	2.24	3.20	4.28	5.40	6.61	7.55	7.87	6.91	5.28	3.54	2.45	1.98	
	RadStd	0.32	0.42	0.54	0.55	0.61	0.41	0.21	0.25	0.36	0.35	0.34	0.27	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+121
(47)	Regional (18 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page